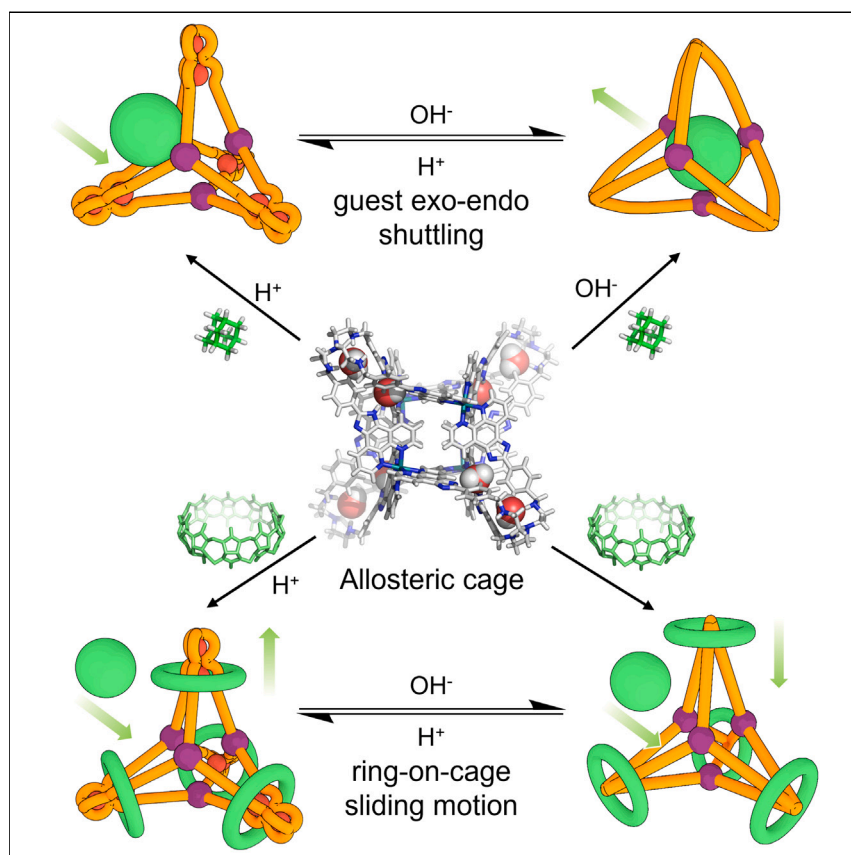


Article

A robust protein-mimicking metallo-amine cage showing proton-driven allostery with water as the effector



We introduce a convenient self-assembly and *in situ* post-modification strategy for the synthesis of a biomimetic cage that combines the advantages of coordination assembly, dynamic covalent-bond assembly, and *in situ* transformation to produce a stable, flexible, and multifunctional protein-mimicking metallo-amine cage, giving rise to a proof-of-concept cage model to imitate the protein allostery for man-made molecular machine and smart biomaterials.

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Highlights

A protein-like metallo-amine cage
enabling the allosteric regulation
by a water effector

A multi-cavity, multi-site, and
multi-valence cage for multimodal
host-guest dynamics

A simple cage model showing
abundant protein-like properties
and complicated functions

Excellent robustness for the acid-
base-regulated guest binding and
drug delivery



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Article

A robust protein-mimicking metallo-amine cage showing proton-driven allostery with water as the effector

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SUMMARY

The allosteric effect is omnipresent in proteins with two distinct binding sites communicating over a distance to control biological functions and complex processes, which drives an everlasting innovation of molecular-scale devices and machines with allosteric functions. Herein, we report a protein-mimicking metallo-amine cage (MOC-68) with excellent tolerance for harsh acid-base conditions, presenting a proof-of-concept example of proton-driven allosteric regulation of multi-state transitions and molecular motions controlled by a water effector as in the natural system. Owing to its multiple N groups for proton transfer, 8 water-binding allosteric sites for remote control, and one endo-cavity and six exo-cavities for guest exchange, the acid-base or solvent switching protocols of cage conformation, charge-state, amphoteric property, phase transfer, guest dynamics, and, unprecedentedly, ring-on-cage sliding and capping/uncapping motions have been demonstrated; this not only provides a unique allosteric cage model for molecular machines and smart biomaterials but also promises potential in drug delivery and release.

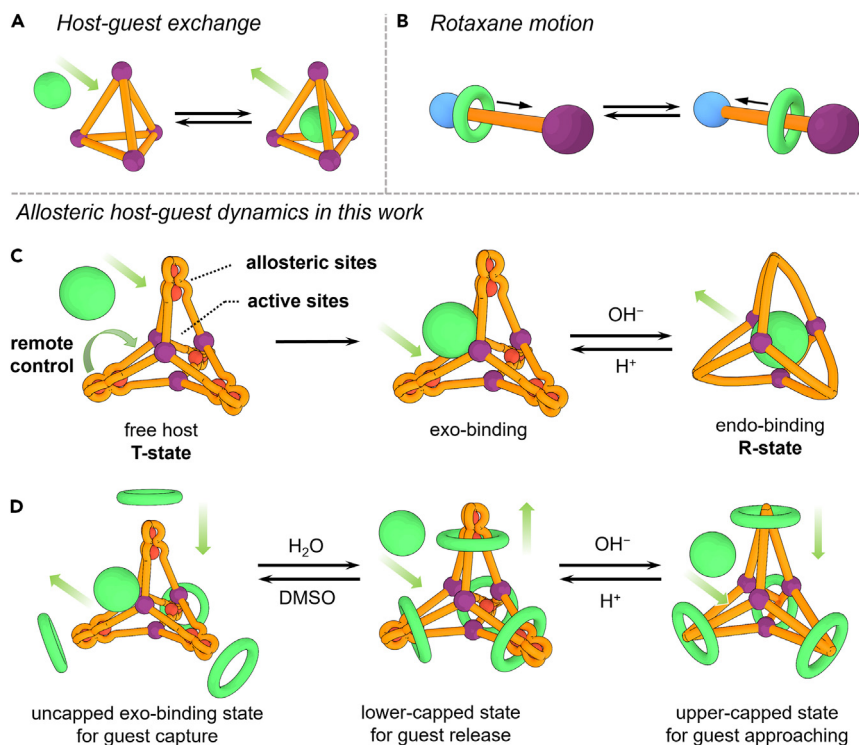
INTRODUCTION

Allostery, referred to as the “second secret of life,”^{1,2} plays the pivotal role in the regulation of protein functions and biological processes in the life system, for example, operating biomolecular machines to undergo energy-driven cycling between various functional and structural states.^{3–6} Hemoglobin, the prototypical allosteric protein as an example, shows conformational changes between the relaxed-state (R-state) with higher O₂ affinity and the tensed-state (T-state) for O₂ release, which can be regulated by H⁺ and CO₂ known as Bohr effect^{7,8} and involve the binding of additional water molecules.² To imitate such a natural mechanism, a great effort has been made for decades to implement allosteric concept into artificial systems to achieve the functions such as molecular recognition,^{9–11} devices,^{12–16} drugs,^{17–19} and catalysis.^{20–22} However, so far, most regulation processes in abiotic systems are accomplished by the aid of anionic and cationic effectors,^{9,10,23,24} rarely achievable by a neutral molecule.²⁵ It is known that the water-protein interactions not only fundamentally underpin protein folding, function, and dynamics but also actively mediate molecular recognition and catalysis.^{26,27} Unfortunately, the vital role of the water effector is heretofore difficult to be experimentally demonstrated and manipulated, and the artificial allosteric system with water molecule acting as the effector remains elusive.²⁸

THE BIGGER PICTURE

The allosteric effect is crucial in the operation of protein machines, igniting much interest in implementing allosteric regulation and mechanisms into artificial systems to achieve biomimetic functions in molecular machines, devices, drugs, and catalysis. The H₂O-protein interactions are ubiquitous in life, but the role of the water effector is difficult to experimentally demonstrate and manipulate. We report a convenient stepwise post-synthetic protocol to construct a flexible cage containing H₂O-binding allosteric sites and multi-guest-binding active sites, presenting protein-like properties such as multi-state conformation, charge tunability, amphoteric property, guest recognition, and phase transfer and achieving acid-base control of various host-guest dynamic motions and movements with water as an effector. This work broadens the possible applications of synthetic cages in harsh conditions, providing wide promise in man-made molecular machines, allosteric catalysis, and drug delivery in the future.



**Scheme 1. Molecular motion modes in different supramolecular systems**

(A) Host-guest exchange showing dynamic ingress and egress of the guest via a cage window.

(B) The ring motion along the rotaxane.

(C) Allosteric host-guest dynamics showing remote control of the guest inclusion mode in either an exo-binding state or the endo-binding state driven by a water effector via the pH swing between the T-state and the R-state.

(D) Allosteric host-guest dynamics showing capping and uncapping of the cage vertices by rings depending on the solvent change, and the ring motion driven by the pH-regulated cage conformation change.

Since an allostery mimicking system needs to integrate distinguishable substrate binding sites for signal communication, it is usually elaborately designed and synthesized through a tedious procedure, for instance, by incorporating at least one allosteric site and one macrocyclic or cage-like receptor.^{9,10,14} By contrast, a self-assembly process, which is ubiquitous in biomolecules, is expected to offer a more convenient and efficient way to construct such a biomimetic system by means of hydrogen bonds or dynamic covalent/coordination bonds. Along this line, some organic and coordination cages have been designed to manifest elegant allosteric properties.^{15,25,29–33} Nevertheless, the cage models based on the dynamic bonds or labile metal centers are often lacking in chemical stability to resist acid and base conditions, thus limiting their modulatory operations under harsh conditions. Moreover, the controllable guest inclusion^{34–37} and directional molecule motion^{16,38–41} for the design of molecular machines have always been a fascinating field in supramolecular modulation, not to mention the devices that combine the properties of both (Scheme 1). However, the synthesis of complex cage structures possessing multiple allosteric sites and multivalent receptors for multi-point molecular motion control is always challenging.^{25,31,42,43}

By virtue of post-modification to transform dynamic imine cages into robust amine organic cages,^{44–47} herein we propose a two-step consecutive post-synthetic

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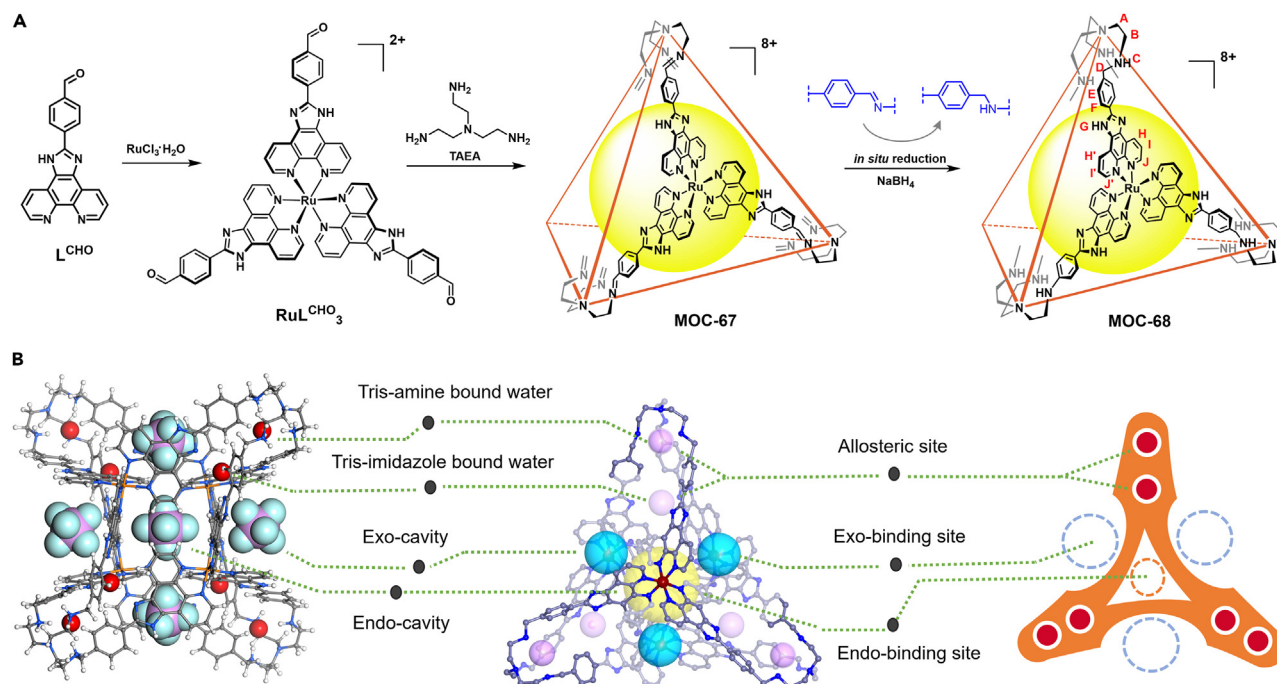


Figure 1. Post-synthetic route and allostery of the protein-mimicking cage MOC-68

(A) Subcomponent self-assembly of the imine cage MOC-67 from inert Ru(II)-metalloligands and *in situ* reduction transformation into robust MOC-68. (B) Crystal structure and representative cartoon of MOC-68 illustrating guest-binding cavities and allosteric sites. The molecules in space-filling mode (PF₆[−]) and the balls/circles in colors demonstrate different exo- and endo-cavities of MOC-68.

protocol (Figure 1) to construct a stable coordination organic cage: (1) subcomponent self-assembly of a tetrahedral metallo-imine cage (MOC-67) based on the dynamic imine bonds⁴⁸ and the thermodynamically stable Ru(II)-metalloligands^{49,50} and (2) *in situ* reduction⁴⁴ of MOC-67 into a metallo-amine cage (MOC-68) characteristic of organic flexibility and stability, as well as metal kinetic inertness. MOC-68 presents some prominent protein-resembling characters unprecedentedly achieved in such a simple supramolecular cage⁵¹: (1) well-organized multiple amine and imidazole (Im) sites for protonation, deprotonation, and water interactions; (2) multivalent charge states from theoretically +36 to −4 tunable by pH variations, showing amphoteric property of isoelectric point and alterable solubility for phase transfer between different solvents; (3) distinct exo- and endo-cavities for guest recognition and movement; and, most importantly, (4) allosteric conformation change allowing distant communication to control multi-state switching of guest dynamics and molecule motions (Schemes 1C and 1D). These biomimetic features are facilitated by excellent cage stability against strong acid and base treatments (pH = 2.8 or 13.5 for 1 month) and visible light irradiation (5 W, 450 nm for 3 days).

RESULTS AND DISCUSSION

Synthesis and characterizations of MOC-68 and (CB[10])₄@MOC-68

The coordinatively self-assembled metal-organic cages usually feature multivalent charges, well-defined shapes, and water solubility, which are beneficial to guest recognition and function design^{52–59} but subject to chemical instability under harsh conditions. By contrast, the error-checking imine bonds facilitate self-assembly and transformation of organic cages with superior flexibility and dynamics^{32,60–62} but usually cause poor solubility in aqueous solution. Our strategy is to combine their advantages while circumventing their drawbacks through a consecutive

post-synthesis as shown in Figure 1. The imine cage MOC-67 is first prepared from a [4+4] self-assembly of the inert metallo-precursors $\text{RuL}^{\text{CHO}}_3$ and the flexible tris-amine TAEA (2,2',2''-triaminotriethylamine), which is then reduced *in situ* to give the stable amine cage MOC-68. Benefiting from the efficient subcomponent self-assembly by the aid of dynamic imine bonds,⁴⁸ MOC-67 is produced quantitatively and directly used as the intermediate for transformation into MOC-68 without tedious isolation and purification, giving a good overall yield of 87%. Moreover, suspension of cucurbit[10]uril (CB[10]) on an aqueous solution of MOC-68 results in further self-assembly of a rare ring-on-cage system, $(\text{CB}[10])_4@ \text{MOC-68}$, where the tetrahedral cage is capped by CB[10] rings on its four vertices (see details in supplemental information and Figures S1–S19). It is worth mentioning that although the cage synthetic strategy based on the reduction of imine to amine has been widely applied, the two-step post-synthetic protocol reported here merits attention in the extendable practice for the assembly of multifunctional coordination organic cages: (1) combining virtues of coordination cages and organic cages, (2) utilizing inert metalloligand to avoid concomitant reduction interference from metal centers, and (3) resorting to reversible and dynamic imine bond to guarantee predictable and controllable cage assembly. These together afford a designable way to achieve cage receptors featuring multi-cavities, multi-sites, multi-valences, and dynamic conformations, meeting the practically demanding properties such as solubility, stability, flexibility, and guest-binding capacity.

The cage structures have been determined by the X-ray single-crystal diffraction, confirming that MOC-68 consists of 4 Ru-metalloligands as the tetrahedron faces and 4 tris-amines as the vertices (Figures 1B and S18; Tables S1 and S2). Each individual cage displays a homochiral $\Delta\Delta\Delta\Delta$ or $\Lambda\Lambda\Lambda\Lambda$ configuration with 4 tris-chelating Ru centers having the same clockwise or anti-clockwise fashion. It is worth noticing that the whole cage is shrunken (T-state) by 8 water molecules bound in the tris-amine cavities and tris-Im cavities on each vertex through H-bonds ($\text{O}\cdots\text{H}\cdots\text{N}$ or $\text{N}\cdots\text{H}\cdots\text{O}$ distances: 2.680–2.756 Å), leading to the formation of a closed central endo-cavity of 7.3 Å diameter surrounded by 8 phenanthroline (Phen) rings, and 6 open portal exo-cavities of 8.5 Å separation folded up by two Ru-metalloligands. The crystal structure of $(\text{CB}[10])_4@ \text{MOC-68}$ unveils an unusual ring-on-cage supramolecular system showing host-capping-host interactions, in which 4 water-tightened vertices of MOC-68 are capped by 4 CB[10] rings (Figures 2E and S19; Video S1).

MOC-68 retains the cage structure in solution, which is evident from the high-resolution electrospray ionization time-of-flight mass spectrometry (HR ESI-TOF-MS) measured in $\text{CH}_3\text{CN}:\text{H}_2\text{O} = 1:1$ (v/v) media. A series of successively charged peaks assignable as $[(\text{MOC-68})^{8+} - m\text{H}^+ + n\text{BF}_4^- + x\text{H}_2\text{O}]^{(8+m-n)+}$ species are observed, well matching with the simulated isotopic patterns (Figures 2C and S17). The ^1H NMR spectra of MOC-68 in CD_3CN present a set of sharp resonances, indicative of a highly symmetric structure with the dynamic radius (r) estimated as 14.5 Å from ^1H -DOSY spectra (Figure 2A), consistent with the single-crystal structure analysis. The signals on Phen moieties split into two sets, suggesting distinct magnetic environments of the protons located inside (H_H , I , J) and outside ($\text{H}_\text{H'}$, I' , J') the cage. Noteworthy, two types of H-binding water molecules are discernible on NMR spectra, where the H_K at 12.2 ppm is correlated with adjacent aliphatic $\text{H}_{\text{A,B,D}}$, amine H_C , and phenyl H_E , whereas the H_G at 17.1 ppm is crosslinked with adjacent Phen H_H and phenyl $\text{H}_{\text{E,F}}$ (Figures 2B and S10–S12) on the ^1H - ^1H NOESY spectra, denoting that H_K is related to the encapsulated water in tris-amine cavities, whereas H_G is related to the encapsulated water in tris-Im cavities. The addition of HBF_4 into the CD_3CN solution of MOC-68 causes the disappearance of H_K and H_G

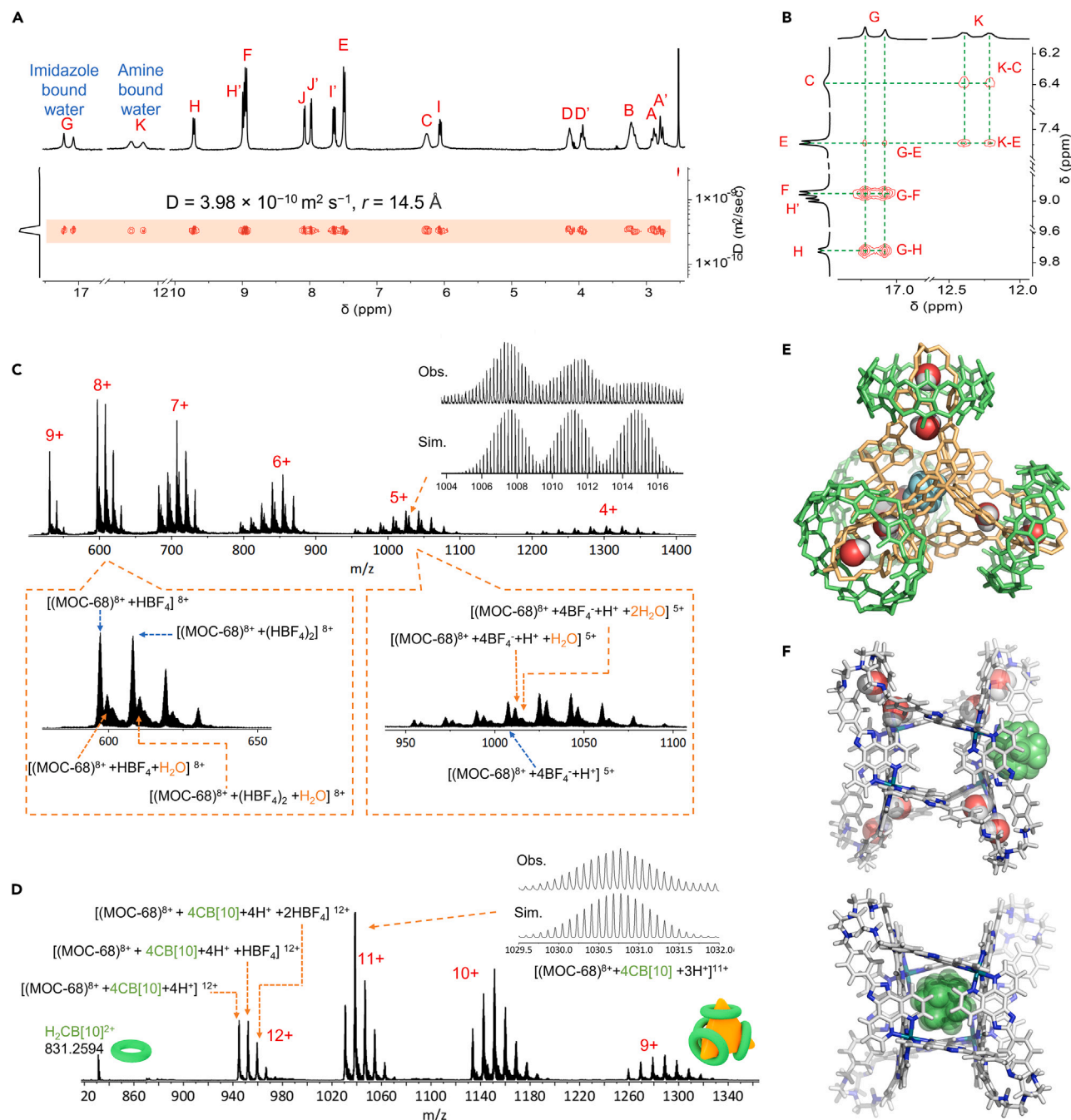


Figure 2. Spectroscopic characterization and single-crystal structure

(A) ^1H NMR and ^1H -DOSY spectra of MOC-68 (400 MHz, CD_3CN , 300 K).

(B) Partial ^1H - ^1H NOESY spectrum of MOC-68 (400 MHz, CD_3CN , 300 K).

(C) HR-ESI-MS spectra of MOC-68 in $\text{CH}_3\text{CN}:\text{H}_2\text{O} = 1:1$ (v/v), pH = 5.8, showing charge series and water-binding peaks.

(D) HR-ESI-MS spectra of $(\text{CB}[10])_4@ \text{MOC-68}$ under neutral conditions in H_2O . Green ring denotes CB[10], and yellow tetrahedron denotes MOC-68.

(E) Single-crystal structure of $(\text{CB}[10])_4@ \text{MOC-68}$. Encapsulated water and PF_6^- shown in the space-filling mode.

(F) Forcite simulation of the exo- and endo-binding modes of the adamantane guest (space-filling) by MOC-68.

See also Figures S10, S17, and S50.

signals, which suggests that H_K and H_G are not simply arisen from the protonation of amines and Ims but consolidated by H-bonds (Figure S15). Such an H-bond-induced low-field chemical shift is often observable in intramolecular H-binding compounds⁶³ but rarely observed in intermolecular circumstances, which is comparable with the short H-bond or low barrier H-bond (LBHB) in the active site of enzymes to serve as a catalytic triad for effective proton transfer reactions.^{64,65} Therefore, observations of these H-binding proton signals are owing to the formation of strong H-bonds between the encapsulated water and tris-amines/tris-Ims, although the definite location on water or amine/Im is hard to determine. This hypothesis is also supported by pH-dependent variation of these H-binding signals in an aqueous solution (Figure S16). In a neutral DMSO- d_6 :H₂O = 1:2 (v/v) solution, these signals persist, implying a relatively slow exchange process between the bound water and the solvent water in the NMR timescale at room temperature. In an acidic condition (pH $\sim$ 2.6), H_K is kept, whereas H_G disappears, probably due to the proton exchange via Im tautomerism. In a weak basic condition (pH $\sim$ 8.3), both H_K and H_G signals disappear, indicating the release or accelerated exchange of water molecules. Therefore, these H-binding peaks can serve as a signal indicator to monitor the water-binding behaviors. The HR-MS measurement in an aqueous CH₃CN:H₂O = 1:1 (v/v) media also provides evidence of the existence of the H-bonded water inside the cage (Figure 2C).

Acid-base modulation of charge state and solubility

Owing to the presence of multiple proton donors and acceptors (totally, 40 N sites on the 4 apical-tripods, including 1 $\times$ 4 tertiary [$\equiv$ N] amines, 3 $\times$ 4 secondary [=NH] amines, 3 $\times$ 4 Im-N as H-acceptors, and 3 $\times$ 4 Im-NH as H-donors) on MOC-68, the charge state of the cage can be readily modulated by altering the solution's pH. Theoretically, the total charge state can escalate to positive +36 through full protonation (28 H-acceptors plus +8 charge of 4 Ru²⁺ centers) in acid conditions and reduce to negative -4 through full deprotonation (12 H donors minus +8 charge of the cage) in basic conditions, leading to a wide charge range to modulate the cage properties such as conformation, polarity, and solubility. Conversion from the metallo-imine cage MOC-67 into metallo-amine cage MOC-68 imparts excellent chemical stability to tolerate such charge variations under harsh conditions. As monitored by NMR spectra (Figures S20–S26), MOC-68 keeps its cage integrity at pD = 2.8 after heating at 80°C for 7 days and remains intact at pD = 13.5 at room temperature for 1 month. Only long-time heating at 80°C under strong basic conditions can damage the cage. This cage also shows super photostability under blue light irradiation and no matter in neutral or acidic conditions. Moreover, it is stable in the presence of an oxidant. After heating in H₂O₂ at 50°C for 24 h, nearly no change is observed from NMR spectra.

The solution dynamics of MOC-68 have been tested by ¹H NMR titration from acid to base by adding NaOH aliquots incrementally to a series of parallel NMR tubes. A constant volume of DMSO- d_6 :H₂O or DMSO- d_6 :D₂O = 1:2 (v/v) aqueous solutions is maintained during titration. Since the dissolving of pristine MOC-68 in water gives a pH value of 5.6, approximately 8 pre-protonated N sites, 20 equiv HBF₄ is loaded at the beginning to ensure full protonation of the cage (Figures 3A, 3B, and S27–S30; Table S4; see details in supplemental information). Plots of pH variations vs. NaOH equivalents can be generally divided into three stages: (1) successive deprotonation of most exchangeable protons in the pH = 2.61–8.31 range with the addition of 0–32 equiv NaOH; (2) precipitation of MOC-68 in the pH = 8.52–11.06 range with the addition of 34–40 equiv NaOH; (3) complete re-dissolution of precipitate in the pH = 11.56–12.85 range with the addition of 42–58 equiv NaOH. The cage precipitates in stage 2 due to deprotonation of Im-NH protons to balance +8 charges on 4 Ru²⁺

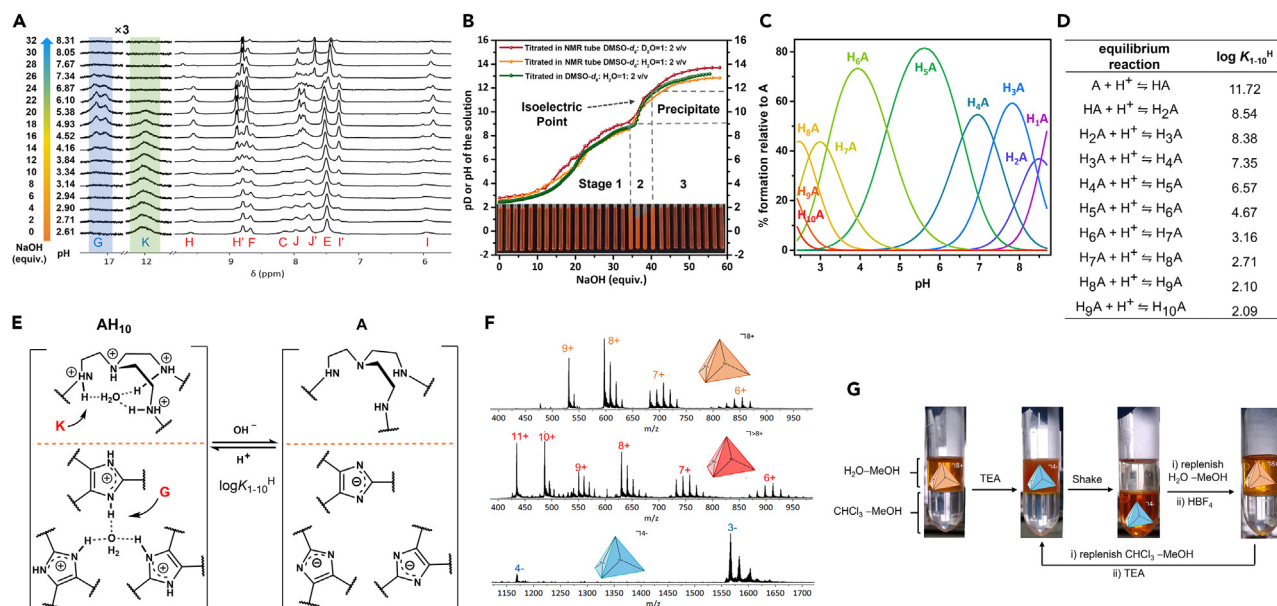


Figure 3. Charge-state and protonation constant of MOC-68

(A) Partial ^1H NMR titrations of MOC-68 from acidic to basic conditions with NaOH (400 MHz, 0.5 mM MOC-68, 10 mM HBF_4 , $\text{DMSO}-d_6:\text{H}_2\text{O} = 1:2$ (v/v), titrated with 100 mM NaOH stock solution by adding NaOH incrementally to NMR tubes while keeping a constant solution volume, 300 K). H_G and H_K signals are highlighted.

(B) Plot of pH or pD values vs. NaOH equivalents added. Curves in different colors show pH or pD variations measured from the individually titrated NMR tubes (reddish and orange) or the continuously titrated solution for protonation constant determination (dark green). The photograph of the individually titrated NMR tubes is demonstrated below.

(C) Species distribution diagram of the protonation of MOC-68 determined by NaOH titration of $\text{MOC-68} \cdot (\text{BF}_4)_6$ (0.5 mM, 10 mM HBF_4 , $\text{DMSO}-d_6:\text{H}_2\text{O} = 1:2$ (v/v), titrated with 50 mM NaOH stock solution, $V = 5$ mL, $I = 25$ mM, $T = 293 \pm 0.5$ K, in N_2 atmosphere).

(D) Stepwise protonation constants of MOC-68 in $\text{DMSO}-d_6:\text{H}_2\text{O} = 1:2$ (v/v) solution.

(E) Overall deprotonation equilibria of the simplified deca-acid of MOC-68 for water H-binding and release.

(F) HR-ESI-MS spectra of MOC-68 under neutral condition (brown tetrahedron) in CH_3CN , acidic condition (reddish tetrahedron) in CH_3CN containing 100 equiv HBF_4 , and alkaline condition (cyan tetrahedron) in CH_3OH containing 100 equiv NaOH (negative ion mode).

(G) Sequential phase transfer of MOC-68 between aqueous (upper) and organic (lower) solutions triggered by acid-base swing.

See also Figures S27, S31, and S33.

centers, leading to a low-charged or neutral state of the cage to reduce the polarity and solubility in aqueous solutions, which is similar to the isoelectric point in protein ionization (Figures 3B and S31). The formation of a -4 negative cage is expected in stage 3 through full deprotonation of 12 Im-NH H-donors, resulting in 12 Im- N^- sites coexisting with 4 Ru^{2+} centers on the cage, which in turn significantly increase the polarity of the cage to improve its solubility (Table S6). ^1H NMR spectra in this stage turn indiscernible at the beginning, but the resonances gradually emerge above $\text{pH} = 12.5$ and can be completely restored by acidifying the solution again (Figures S27 and S29), implying that complete deprotonation of 12 Im-NH may lead to a relaxation of the negative-cage conformation for more freedom and hydration.

The ^1H NMR resonance profiles in stage 1 show distinct changes along stepwise deprotonation, reflecting cage conformational variations as H^+ hydrons are successively abstracted from the cage (Figure 3A). Strikingly, the H-bonding H_G and H_K signals relating to water encapsulation provide explicit information for such conformational changes. From Figure 3A, we can see that H_K presents but H_G cannot be observed at low pH, denoting that protonation of tris-amines facilitates H-bonding with water to retard fast proton exchange,²⁸ whereas H-bonds between water and protonated tris-Ims may subject influence from rapid proton tautomerization on imidazolium rings,

thus becoming silent in NMR spectra. As pH escalates, H_K disappears, whereas H_G emerges gradually in a pH range of 3.8–7.3, suggesting differentiated H-bonding proton exchange for these two types of encapsulated water molecules. Meanwhile, the proton signals on the cage become more complicated in this pH range, indicative of a dramatic cage conformational change. Above pH = 7.6, both H_K and H_G vanish, and the cage proton resonances turn simpler, giving a hint of water release and thereof the formation of a more relaxed cage conformation (R-state). Further evidence come from noticeable variations of proton signals in close proximity to H-bonding water, e.g., H_H and H_I (Figures 3A, S27, and S29).

To further understand the multiple deprotonation states of MOC-68, a more elaborate acid-base pH titration is conducted in the DMSO:H₂O = 1:2 (v/v) solution for the determination of protonation constants (Figures 3B–3E, S31, and S32; Table S5; see details in supplemental information). The titration curve well matches with those obtained from individual NMR tube titration, of which only the portion in the pH = 2.44–8.63 range before precipitation is valid for constant calculation (Figures 3B and S32). The whole cage is simplified as a deca-acid based on its total 40 N sites for proton exchange and the 4-fold symmetry of the cage. A fitting with the Hyperquad program gives 10 stepwise protonation constants $\log K_{1-10}^H$ (Figure 3C) with the protonated cage species distribution represented in Figure 3D. Since deprotonation of such a deca-acid is understandably complex²⁸ and the acidity of every N sites will be unpredictably affected by the high charge state and the electron-withdrawing effect from 4 Ru²⁺ centers,⁶⁶ the exact deprotonation steps are difficult to predict, thus only tentatively proposed (Figures 3E and S32) in mutual corroboration with ¹H NMR resonance variations during titration. The initial deprotonation ($\log K_{10}^H$) may start from the apical tertiary amine owing to the electrostatic and steric repulsion effect within the cryptand cavity,²⁸ which may be followed by the deprotonation of three imidazolium H⁺ hydrons ($\log K_{7-9}^H$) that also subject to electrostatic repulsion in tris-Im cavity. The $\log K_7^H$ – $\log K_{10}^H$ are mainly distributed in the pH < 4 region, in concert with the invisibility of H-bonding H_G signals due to tautomeric imidazolium protons. As the pH increased, deprotonation from Im-NH and protonated secondary amines occurs, corresponding to $\log K_{6-1}^H$, which are even complicated because Im-NH and aliphatic =NH₂⁺ derivatives have a similar pK_a (~10) and they are affected by each other. Nevertheless, since Ru²⁺ coordination is expected to impose more electron-withdrawing effect on Im-NH via conjugation to enhance its acidity,^{66,67} proton release most likely happens at Im-NH sites first, probably alternating with tris-amine sites.²⁸ The most abundant distributions for $\log K_4^H$ – $\log K_6^H$ fall in a 3.5–7.6 pH range, where both H-binding H_K and H_G are observable on NMR spectra. At a pH > 7.6, the $\log K_{1-3}^H$ are predominant in solution, leading to deprotonation of the rest N sites and relaxation of the cage conformation by releasing H-bonding water molecules. The neutral cage is formed at an isoelectric point pI = 10.2, which is calculated from $\log K_2^H$ and $\log K_1^H$ and is well consistent with the value estimated from the precipitation stage (Figures 3B and S31).

HR-MS spectra under acidic, neutral, and basic conditions display distinctive charge series (Figure 3F), further confirming the above charge-state regulation. The higher charge up to 11+ appears in the acidic solution, attributable to the protonated species of MOC-68. Compared with the same charged species in neutral conditions, the peaks are shifted to higher molecular weight, indicating the inclusion of more BF₄[−] anions (Figure S33). By contrast, in a strong basic solution, peaks of −4-charged negative-cage are observed, corresponding to full deprotonation of 12 Im-NH sites. These mass spectroscopic results also verify the structural integrity of the cage under harsh conditions. As expected, dramatic charge variations cause consequent changes in

polarity and solubility to display an amphoteric character. The positive cage is soluble in aqueous solutions under acidic conditions, whereas the negative cage is prone to dissolve in organic solvents under alkaline conditions (Table S6). Therefore, a cyclic phase transfer process can be designed by an acid-base swing between the aqueous phase ($\text{H}_2\text{O}:\text{MeOH}$) and the organic phase ($\text{CHCl}_3:\text{MeOH}$) layered in a tube (Figure 3G). The as-prepared MOC-68 is initially dissolved in the aqueous layer of $\text{H}_2\text{O}:\text{MeOH}$. When excess trimethylamine (TEA) is added, the cage solution becomes turbid due to charge-state change, and the transfer of cages into the lower organic layer occurs with shaking. After replenishing the upper aqueous phase with acid, the cages can transfer back to the aqueous layer. Such a reversible phase transfer can be continued by replenishing the lower organic phase with base, which is useful for applications in molecular transport, separation, and catalysis.^{67,68}

Proton-driven allosteric modulation of host-guest dynamic motions

Based on the above acid-base conditioning of the amphoteric cage flexibility and charge state in concert with the H-bonding water molecules, we further investigate the guest encapsulation behaviors of MOC-68 (Figures 4 and S34–S65). A similar acid-base ^1H NMR titration is performed in the presence of 1 equiv adamantine (ADH) guest (Figures 4A and S34), revealing an allosteric effect on the host-guest dynamics that closely resembles the allosteric proteins with the T-state and R-state of an MWC (Monod-Wyman-Changeux) model.² The insoluble ADH is captured by the cage into an aqueous solution via the phase transfer to shift the guest proton signal upfield (Figure 4D). Noticeably, in the 4.9–7.6 pH region, the signal shifts start to increase obviously along with the addition of NaOH, suggesting a conformation change of the cage to allow a deeper encapsulation of the guest. Simultaneously, the H-bonding H_G and H_K signals gradually vanish, whereas the cage H_H signal concomitantly varies in position, implying exchange acceleration and gradual release of the H-bonded water to relax the cage for closer host-guest interactions. Above pH 7.6, a new guest proton signal evolves at a higher field (–1.2 ppm), indicating the occurrence of an enclosed inclusion of the guest. These observations reveal that the ADH guest can be captured in either exo-cavity or endo-cavity of the cage (Figures 1B and 2F), depending on the flexibility and conformation states of the cage. ^1H NMR guest titrations of MOC-68 testify that only the exo-binding guest is present in the acidic condition (pD = 3.40), showing a binding constant $K_\text{a} = 473.69 \pm 19.9 \text{ M}^{-1}$ calculated from 1:1 isotherm fitting with Bindfit⁶⁹ (Figures S36 and S37). Although in a basic condition (pD = 8.70), both exo- and endo-bindings happen with rather a slow exchange in the NMR timescale (Figures S38 and S39), leading to a $K_\text{a} = 392.52 \pm 29.05 \text{ M}^{-1}$ for the exo-binding guest. ^1H -DOSY spectrum confirms simultaneous exo- and endo-binding of the ADH guest by the cage, and ^1H - ^1H NOESY spectrum unveils prominent host-guest interactions in the endo-binding situation (Figures S40 and S41). As a result, in the presence of H-bonded water in acidic conditions, the cage exists in a T-state to prevent the guest from endo-cavity inclusion, whereas the release of the H-bonded water in basic conditions endows the cage with an R-state to enlarge and open the endo-cavity for the ADH guest ingress, arisen from the negative-charge repulsion between the deprotonated Im^- anions. The guest signal shifting curve well matches up with the decremental distribution of H_4A and H_5A (Figure 4B), revealing that the proton-driven equilibria from H_5A and H_4A to H_3A represent the key step to control the host-guest dynamics (Figures 3E and S32). As H_3A distribution increases, whereas those of H_4A and H_5A diminish, the transformation of cage conformation from the T-state to the R-state and guest motion from exo-binding to endo-binding takes place. When elevating the pH above 8.3, the exo-binding form is completely translated to the endo-binding form, triggered by a complete removal of H-bonded

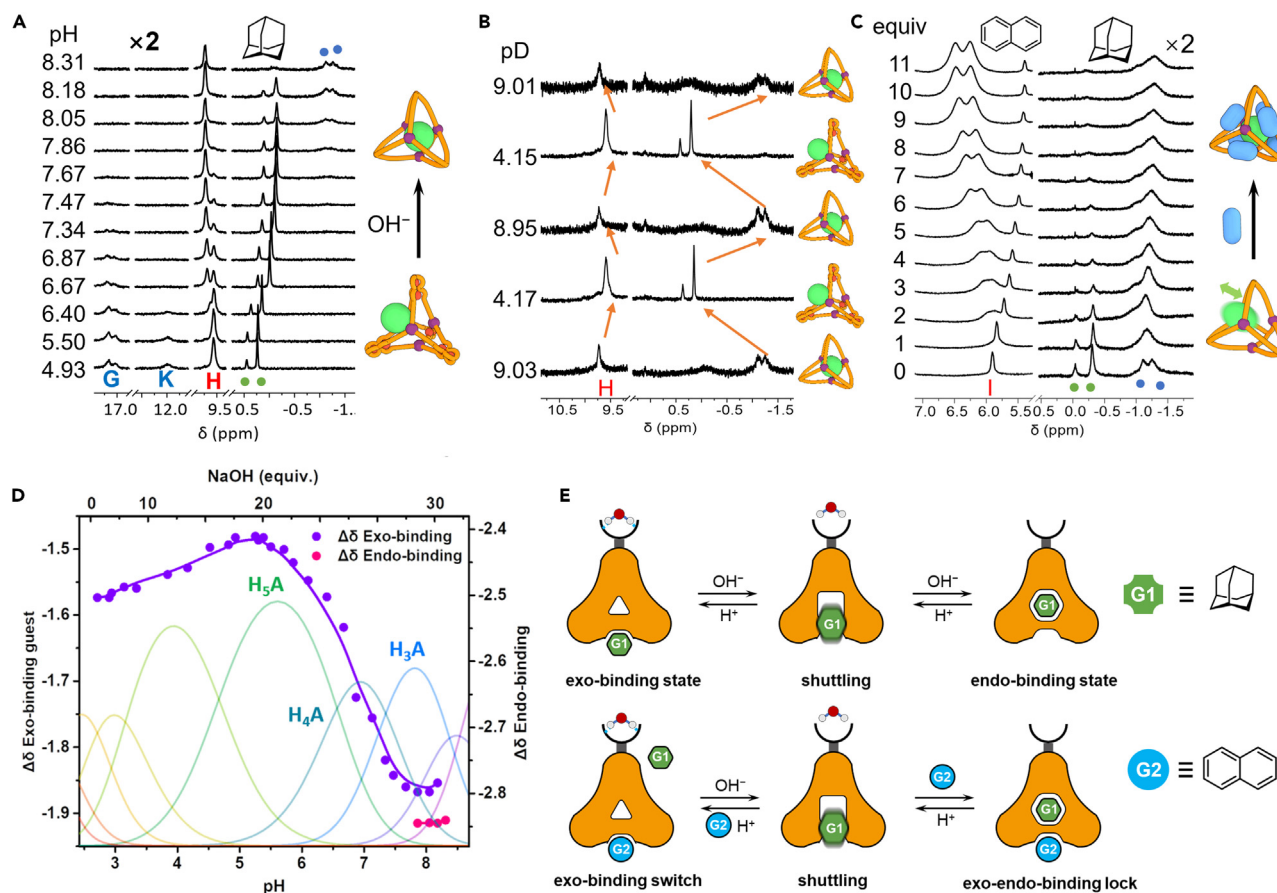


Figure 4. Allosteric modulation of host-guest dynamic motions

(A) Partial ^1H NMR spectra showing pH-regulated exo- and endo-inclusion of the ADH guest by MOC-68 in concert with H-bonded water release. Green dots: exo-binding ADH; blue dots: endo-binding ADH (400 MHz, 0.5 mM MOC-68, 0.5 mM ADH, 10 mM HBF_4 , $\text{DMSO}-d_6$: H_2O = 1:2 (v/v), titrated with 100 mM NaOH stock solution, 300 K). The cartoons represent the host-guest interaction fashions (similar hereafter).

(B) Proton-driven exo-binding and endo-binding switch of ADH guest.

(C) Partial ^1H NMR spectra showing dynamic motions of ADH guest driven by the secondary NAP guest inclusion (400 MHz, 1 mM MOC-68, 1 mM ADH, 10 mM NaOH, $\text{DMSO}-d_6$: H_2O = 1:2 (v/v), titrated with 500 mM NAP stock solution, 300 K).

(D) Chemical shift variation curve of the exo- and endo-binding ADH guest in comparison with the protonated cage species distribution.

(E) Cartoon demonstration of host-guest dynamic motions in the case of one-guest (upper) and two-guest (lower) systems along with the H-bonded water release and cage conformation change (represented by swelling polygons) showing pH driven cycling performance. Only one of the three water effector and guest motion sites is drawn for the sake of clarity.

See also Figures S34 and S35.

water (Figure 4A). As a consequence, a proton-driven allosteric switching protocol may be established with guest in-and-out shuttling motions between exo- and endo-binding sites, which can be cyclically manipulated by the pH swing of the solution (Figures 4B, 4E, and S35).¹² Taking account of the slow exchange process between the exo-binding state ($\text{MOC-68} \supset \text{exo-ADH}$) and the endo-binding state ($\text{MOC-68} \supset \text{endo-ADH}$), the multi-state switch of cage conformations and guest motions is accessible through fine and continuous pH regulation (Figures 4A and 4E), which is characteristic of a signal process via distant allosteric communication, useful as an encodable system with logic gate and device functions.¹⁴

The pH-dependent host-guest dynamics are also observed for the naphthalene (NAP) guest (Figures S42–S45). In acidic conditions, multiple NAP guests obviously undergo a fast exchange in the exo-cavities of a T-state cage, showing little influence

on the cage signals. Although under basic conditions, significant and contradictory shifts of the inner and outer cage protons are observed along NAP titration, suggesting closer host-guest interactions via exo-endo exchanging as the cage is relaxing to R-state. Such significant enhancement of the multiple guests binding is helpful to the allosteric cooperativity,⁷⁰ which is applied to the dynamic regulation of a more complex ADH and NAP two-guest system. Successive addition of NAP guests into an acidic cage solution containing 1 equiv ADH results in wax-and-wane signal variations of the ADH and NAP guests (Figure S46), affording an exo-binding guest switch model ($\text{MOC-68} \supset \text{exo-ADH} \rightleftharpoons \text{NAP}$) with the ADH guest replaceable by the NAP guest (Figure 4E). Although basifying this solution to $\text{pD} = 8.51$ leads to a preferential encapsulation of adamantane again in the endo-cavity due to better host-guest adaptability (Figure S46). Therefore, under basic conditions ($\text{pD} = 8.70$), the addition of NAP guests facilitates ADH guest transition from the exo-binding site to the endo-binding site, leading to a locking state ($\text{MOC-68} \supset \text{endo-ADH/exo-NAP}$) with the ADH guest stuck in endo-cavity, whereas the NAP guests occupying exo-cavities (Figures 4C, 4E, and S47). Such a locked state can be unlocked by acidifying the solution. These host-guest dynamics may provide a simple cage model for the multi-state switching of guest motions through multi-point interactions with multiple allosteric sites on a multivalent receptor.

A more intriguing allosteric regulation model is observed for the ring-on-cage system $\text{CB}[10]_4\text{@MOC-68}$ (Figures 5 and S48–S56). The phase transfer of insoluble $\text{CB}[10]$ into an aqueous solution ($\text{DMSO:D}_2\text{O} = 1:20$, v/v) of MOC-68 leads to the formation of a host-capping-host structural model with 4 vertices of the cage surrounded by 4 $\text{CB}[10]$ rings, which shows rather a slow exchange and is confirmed by HR-ESI-MS and ^1H -DOSY identification (Figures 5B and S48–S50). In acidic conditions ($\text{pD} = 2.90$), a set of neat proton signals of $\text{CB}[10]_4\text{@MOC-68}$ is obtained, indicating that the cage conformations are uniformed to a rigid T-state and fixed by the tight surrounding of 4 $\text{CB}[10]$ rings. When adjusting the solution to a neutral condition ($\text{pD} = 7.10$), drastic signal shifts of H_F , H_E , and H_I are observed, of which H_F is down-field shifted, whereas H_E and H_I are upfield shifted (Figures 5A and S51). This implies that $\text{CB}[10]$ rings can slide away from a lower-capped state to an upper-capped state along the cage vertices (Figures 5C and 5D), moving from a position more shielding H_F atoms to a position more shielding H_E atoms and relaxing the cage conformation with re-orientation of the neighboring Phen planes to shield H_I atoms more. Two well-discriminated resonance profiles on NMR spectroscopy at acidic and neutral conditions indicate two possible ring-on-cage positions on the vertices. Ion-dipole interaction is one of the commonest driving forces in the $\text{CB}[n]$ host-guest system,⁷¹ which renders excellent binding affinity for cations. However, as seen from the single-crystal structure of the lower-capped $\text{CB}[10]_4\text{@MOC-68}$ obtained from the acidic solution (Figures 2E and S19), strong hydrogen bonds are formed between the protonated Im and carbonyl group on $\text{CB}[10]$ downside (Figure 5F), whereas the upper portal of $\text{CB}[10]$ interacts with protonated amines through the ion-dipole interactions. Therefore, the lower-capped state is generated from the interactions of both topside and downside portals of $\text{CB}[10]$ with cage vertices to give a lower energy state (Figures 5D and 5E). As mentioned above, the imidazolium protons are apt to depart before the amine protons on the addition of a base; hence, deprotonation of imidazolium protons breaks hydrogen bonds between $\text{CB}[10]$ and Im motifs, which elevates the energy of the lower-capped state and thereof shifts the ring to the upper-capped state (Figures 5C–5E). By contrast, under the neutral condition, the secondary amines are still protonated; hence, ion-dipole interactions between $\text{CB}[10]$ and protonated amines will stabilize the upper-capped state as the lower energy state. These distinct ring-cage interactions under neutral and

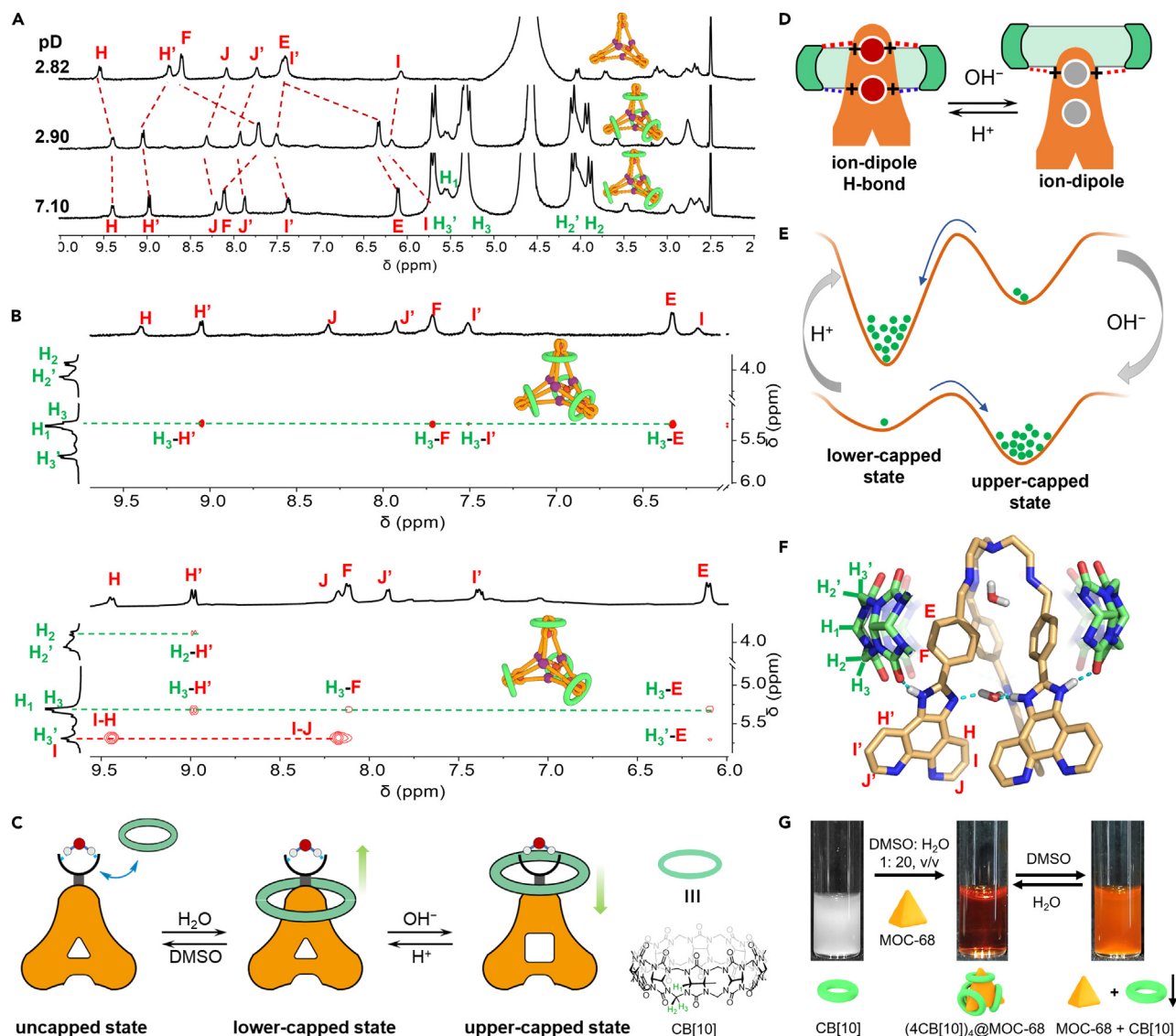


Figure 5. Ring-on-cage sliding motion

(A) ^1H NMR spectra of MOC-68 at pD = 2.82 (upper), (CB[10] $_4$) $_4$ @MOC-68 at pD = 2.90 (middle), and (CB[10] $_4$) $_4$ @MOC-68 at pD = 7.10 (lower) (600 MHz, 1 mM MOC-68, D $_2$ O, 300 K). The cartoons represent the cage and its interaction fashions with CB[10].

(B) ^1H - ^1H NOESY spectra of (CB[10] $_4$) $_4$ @MOC-68 at pD = 2.90 (upper) pD = 7.10 (lower) (600 MHz, 1 mM MOC-68, D $_2$ O, 300 K).

(C) Cartoon demonstration of dynamic motions of ring-on-cage systems along with the cage conformation change (represented by swelling polygons) showing a pH or solvent-driven cycling performance. Only one of the four ring motions is drawn for the sake of clarity.

(D) Cartoon demonstration of the intermolecular driving forces for the sliding motion of the ring-on-cage vertex. Blue dashes denote the ion-dipole interactions, red dashes denote the hydrogen bonds, red balls denote the binding water, and the gray balls denote the exchanging water.

(E) Postulated energy surface diagram of the ring-on-cage system under the acidic and neutral conditions.

(F) Front cross-sectional view of the partial hydrogen bonds based on the single-crystal structure of (CB[10] $_4$) $_4$ @MOC-68.

(G) Illustration of (CB[10] $_4$) $_4$ @MOC-68 formation by adding MOC-68 into turbid solution of CB[10] (DMSO:H $_2$ O = 1:20, v/v), and reversible capping and uncapping motions between CB[10] and MOC-68 driven by solvent change.

See also Figures S49, S52, and S54.

acid conditions are also revealed by the ^1H - ^1H NOESY spectra of (CB[10] $_4$) $_4$ @MOC-68, indicating that CB[10] rings approach $\text{H}_{\text{H}'}'$ and $\text{H}_{\text{I}'}'$ atoms in acidic conditions, whereas stay more remote to only approach to $\text{H}_{\text{H}'}'$ atoms in neutral conditions (Figures 5B, S52, and S54). Further increasing pD to a weak basic condition (precipitate when pD > 8.20) makes the NMR resonances weakened and complicated

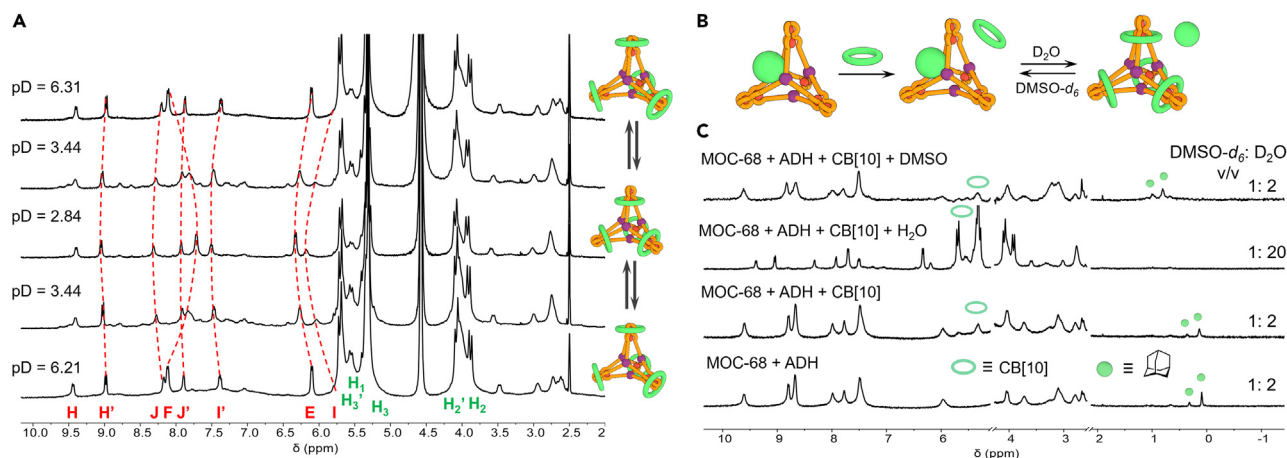


Figure 6. Reversible ring sliding and capping/uncapping motions

(A) ¹H NMR spectral monitoring of the reversible binding state transformation in (CB[10])₄@MOC-68 system swung by pH variation (400 MHz, 1 mM MOC-68, DMSO-*d*₆:D₂O = 1:20 (v/v), 300 K). To avoid the coordination of Na⁺ with CB[10], pH was adjusted with HBF₄ and LiOH instead of NaOH. (B) Cartoon demonstration of CB[10] controlled guest capture-release of the MOC-68 ⊃ exo-ADH system by solvent ratio variation. (C) ¹H NMR spectral monitoring of the reversible guest capture-release of the MOC-68 ⊃ exo-ADH system in the presence of CB[10] (400 MHz, 0.5 mM MOC-68, 20 mM HBF₄, 300 K). Encapsulated ADH signals vanish after the CB[10] is capped on the MOC-68 in DMSO-*d*₆:D₂O = 1:20 (v/v), while reemerge after the ring is uncapped DMSO-*d*₆:D₂O = 1:2 (v/v).

(Figure S51), implying even loose surroundings of CB[10] rings and much relaxation of the cage conformations. Such lower⇌upper sliding motions of the CB[10] rings can be reversibly manipulated by adjusting the solution acidity (Figures 5C and 6A).

In addition, the release of CB[10] rings from the cage vertices can be achieved by changing the DMSO:D₂O ratio from 1:20 to 1:2, owing to a decrease in the hydrophobic interactions between CB[10] rings and the cage,⁷² which is easily observed from the NMR signal change and precipitation of CB[10] (Figures 5G and S56). Taking advantage of the capping and uncapping convertible motions, the guest capture and release can be also achieved by modulating the mechanical capping state (Figures 6B and 6C). At lower water concentrations, the cage is not capped by CB [10]; hence, the ADH guest can be encapsulated at the exo-binding site of the cage. When increasing the water ratio, strong interactions between CB[10] and MOC-68 render the formation of the lower-capped complex and squeeze out the exo-binding ADH. Such guest capture and release behaviors can be reversibly manipulated by adjusting the water content in the aqueous solution. Therefore, this ring-on-cage (CB[10])₄@MOC-68 system provides a mechanical model that is not only able to move the CB[10] rings along the cage vertices, switchable between the lower-capped state and the upper-capped state, but also able to capture and release the guest molecule, switchable between the uncapped state and the lower-capped state with the upper-capped state being ready for guest approaching (Figures 5C and 6B; Scheme 1). Such swing processes can be adjusted by varying the pH value or the solvent to drive sliding and capping motions of the rings in line with the conformational changes of the cage triggered by the water effector, providing potentials in the molecular machine and drug-delivering applications.^{73–75}

pH-triggered controllable drug delivery

Inspired by the above acid-base regulatable guest capture properties, we have tested the inclusion capacity and binding behavior of MOC-68 for some drug molecules and

bioactive guests as a demonstration of controllable drug delivery and release triggered by pH variations (Figures S57–S66). MOC-68 exhibits a satisfying binding ability to hydrocarbon-containing drugs such as camphor, geraniol, L-menthol, and ibuprofen. Based on the mole ratio method analyses of the ^1H NMR titration data, the loading capacities of MOC-68 for these drug guests are centered in a range of 3–6 at a pH = 8.54 (Figure S61), suggesting guest binding in both exo- and endo-cavities of the R-state cage. Therefore, the Hill function analyses were performed to estimate the binding affinity of these multi-guest systems based on the host proton variations, affording the apparent associate constants in a range of $140\text{--}577\text{ M}^{-1}$ which are typical for fast exchange (Figure S62). For camphor, two distinct guest resonance regions are observed along guest titration, of which the proton signals in the higher field are unchanged, whereas those in the lower field show a continuous shift toward the free state (Figure S57). This observation indicates that the exo-binding and endo-binding guests undergo a relatively slow exchange on an NMR timescale, resembling adamantane binding behavior in the basic condition, whereas the more exo-binding guests exhibit a fast exchange in the solution. For the rest of the three drug molecules, only one set of guest signals is observed, which displays a continuous low-field shift toward the free form along with the addition of guest molecules (Figures S58–S60), implying a fast guest exchange among the exo-binding, endo-binding, and free states. Noticeably, when adjusting the host-guest solution from pH = 8.54 to a slightly acidic condition of pH = 6.16, and even lower, remarkable changes in the guest signals happen (Figures S63–S66). The endo-binding camphor resonances decrease and disappear, whereas the coalesced guest resonances of geraniol, L-menthol, and ibuprofen are significantly low-field shifted toward the free states. Moreover, the broad guest signals become well resolved in all cases in acidic conditions. These results indicate that the guest binding in acidic solutions turns weaker and more freely, owing to the transition of cage conformation from the R-state to the T-state to move the endo-binding guest to the exo-binding form, thus facilitating a faster exchange in the solution to be ready for guest dissociation. Such pH-dependent guest-binding events obviously benefit a controllable drug transport and release process *in vivo*, implying a potential to utilize the proton-driven allostery of the cage receptors for the pH-triggered drug delivery.

Conclusions

A convenient stepwise post-synthetic protocol is demonstrated to construct an exceptionally stable and robust coordination organic cage, which takes advantage of metal inertness and organic flexibility with the aid of dynamic imine bond self-assembly to broaden practical applications of the synthetic cages in harsh conditions. Integration of multiply proton donor and acceptor N sites, exo-, and endo-binding cavities, especially functional Im fragments, into a well-organized tetrahedral cage provides a conceptual protein-like cage model to imitate water-protein interactions and allosteric effect that ubiquitously exist in natural biological systems. Multivalent cage conformations can be simply regulated through multiple-point interactions by virtue of a proton-driven water effector, endowing the cage with versatile allosteric functions to control host-guest dynamics via distant communication. Not only the amphoteric properties in relation to charge-state, polarity, solubility, and phase transfer but also the host-guest properties regarding guest recognition, capture, release, and, especially, molecular movements could be precisely modulated by multi-state switching through pH or solvent variations. This work may shed light on how to design and synthesize easy-made artificial supramolecular counterparts of proteins to fundamentally advance biomimetic allostery to a more realistic molecular machine, device, transport, and smart biomaterial applications.

EXPERIMENTAL PROCEDURES

Resource availability

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Cheng-Yong Su (cesscy@mail.sysu.edu.cn).

Materials availability

All materials generated in this study can be obtained upon reasonable request from the corresponding author with a valid materials transfer agreement.

Data and code availability

The accession numbers for the crystallographic data reported in this paper are Cambridge Crystallographic Data Centre (CCDC): 2193725 for MOC-68 and CCDC: 2193724 for (CB[10])₄@MOC-68. Copies of these data can be obtained free of charge from the CCDC via www.ccdc.cam.ac.uk/data_request/cif. Correspondence and requests for materials should be addressed to C.-Y.S.

Synthesis and physical characterizations

Full experimental procedures can be found in the [supplemental information](#).

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.chempr.2023.03.019>.

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AUTHOR CONTRIBUTIONS

C.-Y.S. conceived and directed the project. Y.-L.L. carried out most experiments. X.-D.Z. synthesized CB[10]. C.-Y.S. and Y.-L.L. wrote the manuscript. Y.-H.Q., J.-Q.S., Y.-H.H., C.-H.L., J.-J.C., H.-S.X., and M.P. helped in experiments and data analyses.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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